

Habitable Exoplanet Discovery Framework

Abstract

The detection of habitable exoplanet is crucial so as to understand the possibility of life beyond Earth and the physical conditions prevailing in remote planetary systems. This study makes use of a dataset acquired from the NASA Exoplanet Archive, comprising 5599 confirmed exoplanets as of 12 March 2024. A comprehensive data analysis is conducted to investigate the underlying distributional characteristics of the dataset. Extensive data preprocessing is carried out, involving the application of various imputation techniques selected based on the statistical distribution of each feature and the proportion of missing values. Additionally, new features are derived from existing parameters to enhance the representation of critical attributes influencing classification performance. For the classification purpose, machine learning models namely Random Forest, and XGBoost, are applied which yields high accuracies. Each model is initially trained on all the features to establish baseline performance, followed by training using only the derived features to facilitate comparative evaluation. The results indicate that the XGBoost model, when trained on the complete feature set, achieves the highest accuracy. The insights derived from this study contribute to improving existing models for handling large exoplanetary datasets and support further exploration of exoplanet–stellar environment interactions.

Keywords: *exoplanet, habitability, feature imputation, machine learning, Random Forest, XGBoost*

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Chapter 1

Introduction

The exploration of planets beyond our solar system that could potentially host life is a rapidly evolving and deeply significant area of astrophysics [1]. Any planet that is beyond our solar system is referred as an exoplanet. The discovery of exoplanets, which began in the early 1990s and progressed thereafter [2], has provided more insights to search for the possibility of extraterrestrial life—leading to the confirmation of 5867 exoplanets as of 25 March 2025 [3]. For an exoplanet to be considered habitable, it must be situated within the circumstellar habitable zone (HZ) of its host star. The “habitable zone” (HZ) [4], also called as the “circumstellar habitable zone” (CHZ), is the orbital distance of a planet from its central star at which liquid water could exist on orbiting planets’ surfaces (an important criteria for habitability), provided adequate atmospheric pressure exists [1][4][5]. The HZ is also called the “Goldilocks zone” [6], as it is neither too hot, nor too cold, but just right for life to potentially emerge. The boundaries of this zone are primarily determined by the stellar radiation an exoplanet receives and Earth’s position in the Solar System [7].

Kopparapu et al. (2013) [8] enhanced the conventional definition of the habitable zone by adjusting its boundaries for various stellar types, offering a more precise framework for evaluating exoplanet habitability across diverse stellar environments. In addition to the distance of a planet from its host star, other factors such as atmospheric composition, surface pressure, axial tilt, and magnetic field strength are crucial in determining its potential to support life [9]. Although these advancements have greatly enhanced the theoretical understanding of planetary habitability, observational challenges still remain.

One such significant limitation is the inherent detection method biases, which distorts the distribution of identified planets. Most of the exoplanets are discovered by the transit method, leading to a notable bias towards detecting the exoplanets having shorter orbital periods. This over represents “Too Hot” planets and are closely located to their host stars [10]. Likewise, the Radial Velocity (RV) method tends to detect massive planets with significant gravitational effects on their host stars, leading to a dataset skewed towards large, Jovian planets [11].

To overcome these challenges, recent research [5][8] have started utilizing advanced statistical methods, including machine learning (ML), to improve the habitability predictions and reduce detection biases. ML algorithms can process larger amount of data in a shorter time period by identifying the patterns quickly which may be a tedious task to detect using the traditional methods. As a result, this approach improves the ability to categorize planets according to their potential habitability.

This study presents an in-depth analysis of about 5599 exoplanets that are confirmed and listed in the NASA Exoplanet Archive as of 12 March 2024 [3], with a focus on potentially habitable exoplanets. Unlike the sophisticated methods proposed by Kasting et al. (1993)[5] and Kopparapu et al. (2013)[8] - this study focuses on developing a custom habitability index. The approach involves applying various imputation techniques to handle the missing data. Then, new features are derived from the existing features namely, Earth Similarity Index (ESI), Atmospheric Retention (AR) and Long Term Stability (LTS) which will be instrumental in determining the custom “Habitability Index.” XGBoost classifier is then trained on the processed dataset for the classification purpose.

Furthermore, by carefully evaluating the limitations of the data and methodologies employed, this study emphasizes the necessity of future observational missions with enhanced sensitivity and broader coverage to more accurately determine the prevalence of habitable worlds. Specifically, missions equipped with advanced imaging and spectroscopic techniques could mitigate current detection biases, providing deeper insights into exoplanetary atmospheres and surface conditions.

The organization of the subsequent chapters is outlined as: Chapter 2 provides a literature survey to review the similar work along with the research gap, problem statement, objectives, and scope. Chapter 3 outlines a detailed explanation of the methodology adopted, including the various imputation strategies, deriving the new features, followed by the XGBoost classifier and the feature importance analysis. Chapter 4 presents the findings, the experimental results, analysis tools and the performance of the model. In Chapter 5, the conclusion of this work is discussed by providing brief summary and critique of the findings with scope for future work.

Chapter 2

Literature Review

2.1 Survey of existing system

Planetary habitability refers to the capacity of a planet or moon to sustain life, influenced by factors such as liquid water, atmospheric stability, temperature, magnetic fields, and geological activity [1]. The concept of habitable zone is—the circumstellar region proximal to a star where conditions may support liquid water—is central to identifying potentially life-supporting planets. Other contributing factors include stellar type, planetary mass, and chemical composition.

Kasting [5] proposed one of the earliest models to estimate habitable zones (HZ) around main-sequence stars, using a 1D climate model for planets with $\text{CO}_2\text{-H}_2\text{O-N}_2$ atmospheres. They determined the inner edge of the circumstellar habitable region based on water loss induced by photodissociation phenomena, and the outer edge on CO_2 condensation, laying the groundwork for future HZ calculations.

Later, Kopparapu [8] updated these HZ boundaries using improved atmospheric absorption data and a refined radiative-convective model. Their estimates placed the inner edge at 0.99 AU and the outer edge at 1.70 AU for our Solar System. They also emphasized stellar flux over equilibrium temperature as a better metric for habitability and provided equations applicable across a range of stellar types.

Ramirez [12] proposes an expanded definition of the habitable zone (HZ), arguing that the classical $\text{CO}_2\text{-H}_2\text{O}$ -based model may exclude potentially habitable planets. The study emphasizes the role of additional greenhouse gases such as CH_4 and H_2 , which can broaden the outer edge of the HZ. Using analogs like early Mars and Titan, the paper supports a more flexible, multi-parameter approach to defining habitability, improving target selection for life-detection missions.

Another approach to evaluating potential habitability is the Earth Similarity Index (ESI) [13], which quantifies how Earth-like a planet is on a scale from 0 to 1. On the basis of the parameters like escape velocity, density, radius, and surface temperature. For exoplanets, due to limited data, the ESI often uses stellar flux and either radius or mass. However, its accuracy is limited by uncertainties and assumptions—especially in estimating surface temperatures, which are typically equilibrium values that may not reflect true conditions. Thus, while useful for comparison, the ESI does not directly measure habitability.

Tianqi Chen [14] introduced XGBoost in 2016, which is an effective and scalable execution of gradient boosting machines. The authors developed a novel sparsity-aware algorithm to handle sparse data and introduced an approximate tree learning approach utilizing a weighted quantile estimation method. These innovations significantly enhanced the speed and performance of machine learning models, leading to XGBoost’s prevalent use in both benchmark data science challenges and practical applications.

2.2 Research Gap

Existing models for planetary habitability largely focus on ideal physical and orbital parameters, often overlooking critical factors like atmospheric retention, long-term stability, and planetary evolution. Indices such as the Earth Similarity Index are useful for comparison but fall short in assessing true habitability due to limited data and simplified assumptions. Moreover, current frameworks rarely integrate advanced machine learning techniques for effective classification of exoplanets. There is a clear need for a more comprehensive, data-driven approach that combines astrophysical reasoning with intelligent systems to evaluate and categorize exoplanet habitability more accurately.

2.3 Problem Statement

The growing threat of climate change, environmental degradation, and resource exhaustion has heightened the urgency to identify alternative habitable planets for potential human colonization. While extensive astronomical data on exoplanets is available, the absence of a unified and scientifically grounded framework for evaluating planetary habitability presents a significant challenge. Propose a systematic framework for assessing planetary habitability through a custom Habitability Index that integrates the Earth Similarity Index (ESI), long-term stability, and atmospheric retention. Employ statistical methods or machine learning models to classify exoplanets into potentially habitable, marginally habitable, and non-habitable categories, supported by effective visualizations and scientific reasoning to guide decision-making.

2.4 Objectives

Based on the study of the existing literature and the identified research gaps, the following research objectives are as follows:

- 1) To clean and pre-process the raw data by addressing the noisy and null values.
- 2) To identify the interdependencies among variables and visualize the correlation patterns.
- 3) To derive meaningful features such as the Earth Similarity Index (ESI), long-term stability, and atmospheric retention that might be useful for the custom “Habitability Index”.
- 4) To build a multi-class classifier to categorize planets into 3 classes - Potentially Habitable, Marginally Habitable and Non-Habitable.

2.5 Scope

By applying machine learning models, this study aims to predict whether the exoplanets are habitable or not. Key components include imputation techniques, feature engineering, classification and performance evaluation metrics. The future endeavors may involve exploring absorption coefficients and addressing biases introduced by detection methods to ensure a more balanced and reliable assessment of exoplanet habitability for further refinement. Through this project, the aim is to contribute to the advancement of predictive analytics in space.

Chapter 3

Proposed System

3.1 Framework

Figure 1 explains the workflow of the exoplanet discovery framework.

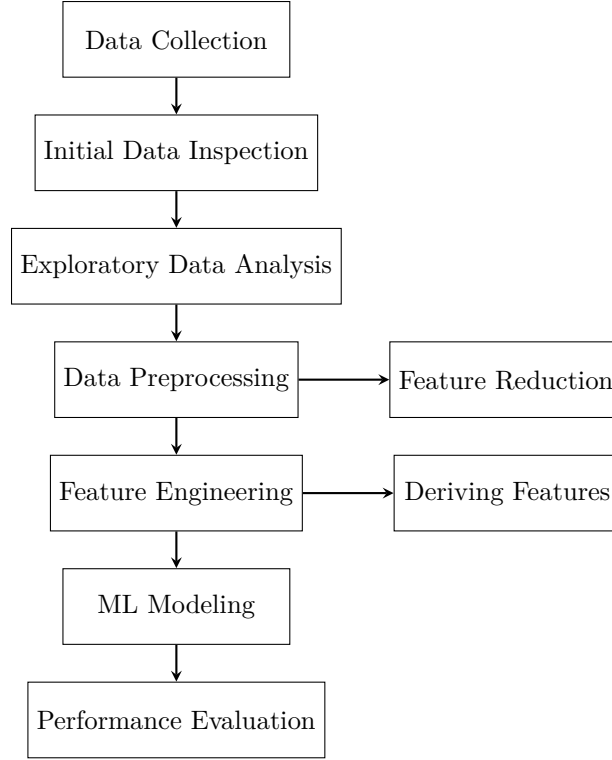


Figure 1: Workflow of the exoplanet discovery framework

The dataset incorporated in this work is retrieved from the NASA Exoplanet Archive [3] which consists of 57 features and 5599 confirmed exoplanets as of 12 March 2024. Subsequent to data collection, an initial inspection is performed to gain insights into the raw dataset, facilitating the identification of potential issues and guiding further preprocessing steps. Next, feature-wise analysis is conducted using various visualization techniques to examine data distribution patterns, detect outliers, and assess the presence of missing values. After thorough data analysis, the initial step in the preprocessing phase involved removing features identified as irrelevant or redundant with respect to the target variable. Following this, appropriate imputation techniques are applied to address missing values. After pre-processing, feature engineering is carried out to construct new features from the existing features, enhancing the classification capability of the models. Thereafter, machine learning models, including Random Forest [15] and XGBoost [14], are implemented and compared based on their performance metrics [16] such as recall, precision, and F1-score. Finally, the models are evaluated using balanced accuracy [17].

3.2 Data Pre-processing

3.2.1 Dataset

This study focuses on the 5599 confirmed exoplanets consisting of 57 attributes, listed in the Planetary Systems Composite Data section of the NASA Exoplanet Archive [3] as of 12 March 2024. The dataset is structured to support the efficient application of computational analysis and the interpretation of complex patterns. Key observational parameters, including exoplanetary orbital characteristics and host star radiative properties, are systematically incorporated. This approach facilitates the extraction of statistically significant relationships, primarily based on fundamental observables such as exoplanet’s surface temperature, flux, radius and the semi-major axis. A summary of the dataset is provided in Table 1 where class 0, class 1, and class 2 represents the count of non-habitable, potentially habitable, and marginally habitable exoplanets respectively.

Table 1: Dataset description

Dataset Source	Attributes	Class 0	Class 1	Class 2	Total Records
NASA Exoplanet Archive	57	5529	29	41	5599

3.2.2 Data Preprocessing Flowchart

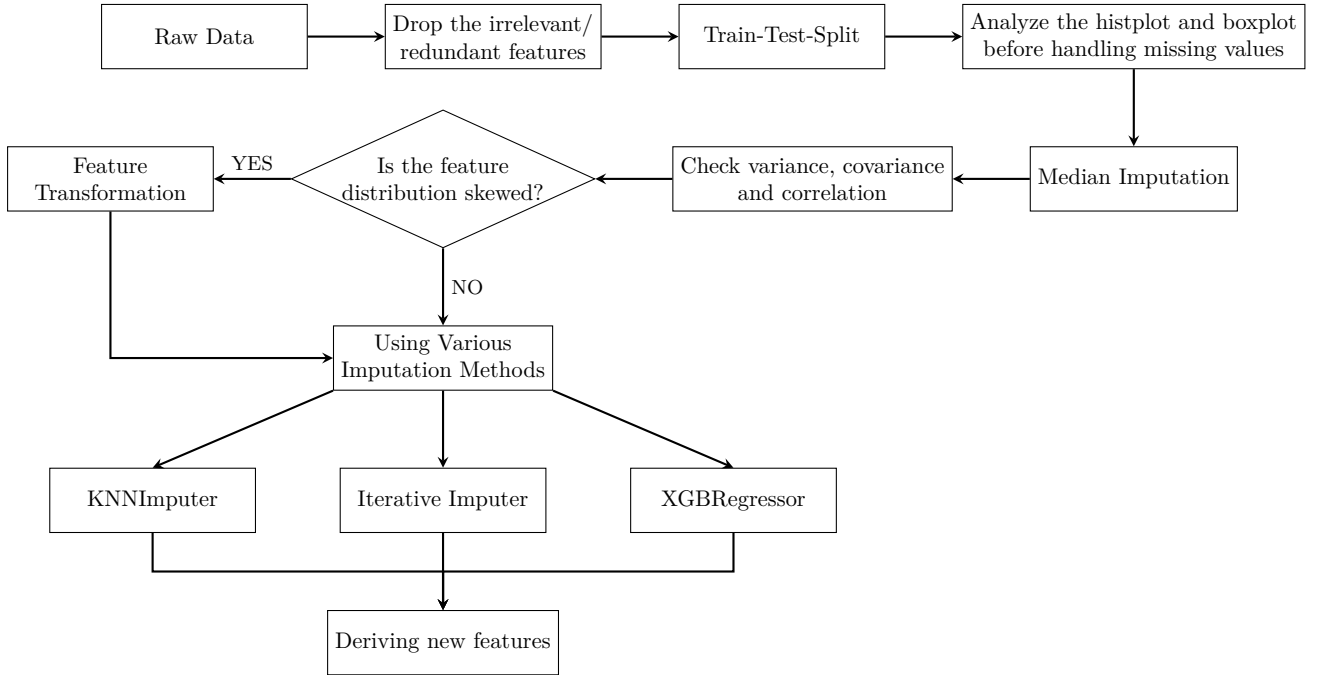


Figure 2: Data Preprocessing Flowchart

Figure 2 describes the steps involved in data preprocessing phase.

3.2.3 Data Imputation Techniques

Data imputation refers to the technique of estimating and substituting missing values within a dataset to maintain its completeness and usability. In real-world scenarios, datasets often contain gaps due to various factors such as measurement errors, data

corruption, or incomplete data acquisition processes. Addressing these missing values is essential, as many analytical and machine learning methods require a complete dataset to function properly. Imputation helps mitigate the negative impact of missing data by using existing information to generate reasonable estimates, thus preserving the dataset's structure and statistical properties. It is typically applied when the amount of missing data is not excessive and when the missingness exhibits a pattern that can be inferred from the available data. In this study, the following imputation techniques are utilized to handle the missing values.

1) Median Imputation

Median imputation is a simple yet robust technique that involves replacing the NaN values by imputing the median of the existing observations in a feature. This method is particularly effective when the data distribution is skewed or contains outliers, as the median is less sensitive to extreme values than the mean. It is commonly applied during initial preprocessing when a quick and reliable estimate is needed without introducing significant bias. Median imputation ensures that the central tendency of the dataset is maintained, making it suitable for numerical attributes where preserving data integrity is crucial while mitigating the influence of anomalies.

2) KNN Imputation

KNN imputation [18] is a distance-based method that estimates missing values by identifying the k most similar instances (neighbors) in the dataset based on feature similarity. The missing value is then imputed using the average (or weighted average) of the corresponding values from these neighbors. This technique is advantageous when the dataset contains local structure or clustering tendencies, as it retains the natural variability within the data. KNN imputation is suitable for both numerical and categorical features and is most effective in moderately sized datasets where computation is manageable and similarity between samples can be meaningfully quantified.

3) Iterative Imputation

Iterative imputation [19] predicts the missing values of a feature using relationships inferred from the other features through a series of regression steps. It performs multiple rounds of estimation, refining the imputed values at each iteration until convergence. This method is particularly useful when there are strong interdependencies among features, allowing it to capture complex relationships that simpler techniques may overlook. Iterative imputation is beneficial in structured datasets with multivariate missing patterns, offering improved accuracy by leveraging the full feature set during imputation.

4) XGBRegressor

XGBRegressor [14], a tree-based ensemble method from the XGBoost library, can be effectively used within an iterative imputation framework to predict missing values. It models the missing feature using gradient-boosted decision trees trained on the remaining observed features. This approach is highly effective in datasets with non-linear relationships and high dimensionality, as XGBoost can model feature interactions and handle various data complexities. Using XGBRegressor as an imputer offers a powerful and flexible alternative to traditional statistical methods, often resulting in more accurate and reliable imputations, especially in large and complex datasets.

3.2.4 Feature Engineering

This study primarily aims to enhance both the accuracy and interpretability of the habitability classification model. Thus, feature engineering is performed by deriving three composite features from the existing physical and orbital properties of exoplanets, namely Earth similarity index, long-term stability, and atmospheric retention. With these three engineered features, the aim is to:

- 1) Reduce dimensionality by focusing on scientifically significant variables, and
- 2) Align the model more closely with the real-world criteria in the search for habitable exoplanets.

1) Earth Similarity Index (ESI)

The (ESI) provides a numerical measure used to evaluate how closely an exoplanet resembles Earth according to its physical attributes. In case of exoplanets, it constitutes factors such as planetary radius and stellar flux. The ESI value ranges from 0 (no similarity) to 1 (identical to Earth). A higher ESI implies that the exoplanet has Earth-like features, which are considered favorable for the development and sustenance of life and improve the chance of it being habitable [20]. The formula used to calculate ESI is as follows:

$$ESI = 1 - \left(\sqrt{\frac{1}{2} \left[\left(\frac{S - S_{\oplus}}{S + S_{\oplus}} \right)^2 + \left(\frac{R - R_{\oplus}}{R + R_{\oplus}} \right)^2 \right]} \right)^2 \quad (1)$$

where, S = Stellar flux received by exoplanet, $S_{\oplus}$ = Stellar flux of Earth,
 R = Radius of the exoplanet, $R_{\oplus}$ = Radius of Earth.

2) Atmospheric Retention (AR)

Atmospheric Retention refers to a planet's ability to hold and maintain a stable atmosphere over extended periods. A stable atmosphere is vital for habitability. It plays multiple roles, including regulating surface temperatures, shielding the planet from harmful cosmic and ultraviolet radiation, and enabling essential biochemical processes. Planets that cannot retain an atmosphere are unlikely to support or sustain life[3]. This characteristic is primarily determined by the surface temperature of exoplanet and escape velocity which is necessary to prevent essential atmospheric gases from dissipating into space. The formula used to calculate Atmospheric Retention is as follows:

$$AR_{\text{ratio}} = \left(\frac{v_{\text{escape}}}{v_{\oplus, \text{surf}}} \right) \times \left(\frac{T_{\oplus, \text{surf}}}{T_{\text{surf}}} \right) \quad (2)$$

where,

AR_{ratio} is the Atmospheric Retention Ratio — a unitless indicator of atmospheric stability relative to Earth,

v_{escape} = escape velocity of the exoplanet,

$v_{\oplus, \text{surf}}$ = escape velocity at Earth's surface,

T_{surf} = surface temperature of the exoplanet,

$T_{\oplus, \text{surf}}$ = Earth's surface temperature.

3) Long Term Stability (LTS)

Long-term stability (LTS) assesses whether a planet’s orbit and environmental conditions are dynamically stable over geological timescales. This measure is essential for evaluating a planet’s potential to sustain life over billions of years, as drastic orbital changes or stellar disruptions can make long-term habitability impossible. It incorporates parameters like the exoplanet’s orbital semi-major axis, which influences orbital period and energy received, eccentricity which affects climate variability due to distance fluctuations, and luminosity of the host star, which determines the energy output affecting the planet’s thermal stability. In this study, LTS is measured on a normalized scale from 1 to 100, where higher values indicate greater orbital stability and consequently higher potential for habitability[3]. Thus, high long-term stability is a strong indicator of habitability.

$$LTS_{\text{ratio}} = (1 - \text{Eccentricity}) \times \left(\frac{L_{\odot}}{L_*} \right) \times \left(\frac{a_p}{a_{\oplus}} \right) \quad (3)$$

where,

LTS_{ratio} is the Long Term Stability ratio — a unitless indicator of the planet’s orbital stability,

$L_{\odot}$ = Sun’s luminosity ,

L_* = luminosity of the exoplanet’s host star,

a_p = exoplanet’s orbital semi-major axis,

$a_{\oplus}$ = Earth’s orbital semi-major axis.

3.3 Classification and Modeling

The approach involves employing two machine learning techniques to classify exoplanets based on their habitability: XGBoost and Random Forest. Both models are ensemble-based and rely on decision tree, which can learn complex relationships and feature interactions that are non-linear in nature. An important advantage of using these tree-based models is that they do not require feature standardization or normalization, making them well suited for datasets with features on different scales.

1) Random Forest

This machine learning model improves predictive performance by building multiple decision trees and aggregating their outputs—using majority voting for classification tasks and averaging for regression. Since Random Forest models are based on decision trees, they can naturally handle non-linear relationships and do not require feature scaling or standardization. It works well with high-dimensional data and can handle missing values effectively. For classification, Random Forest uses majority voting [15].

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_N(x)\} \quad (4)$$

where,

h_i corresponds to the prediction from the i^{th} decision tree,

N is the overall count of trees

The final prediction $\hat{y}$ is determined by the most frequently occurring class (majority vote).

2) XGBoost

XGBoost, [14] a robust and high-performance gradient boosting framework, differs from Random Forest by building trees one after another, where each subsequent tree focuses

on correcting the mistakes of the previous ones. It optimizes a custom loss function using gradient descent. It is well-suited for tabular data, and like Random Forest, it handles different feature scales without requiring standardization. It is known for its speed, accuracy, and tunable hyperparameters, which allow for fine control over model complexity. The objective function used in XGBoost is:

$$\mathcal{L}(\theta) = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (5)$$

where,

$l(y_i, \hat{y}_i)$ represents the loss function,

$\Omega(f_k)$ serves as a regularization component to control model complexity.

K being the number of trees used in the model.

3.4 Performance Evaluation Metrics

- 1) **Recall.** It quantifies the ratio of true positives to the total number of actual positive cases[16][21]. The following formula is used for computing recall:

$$Recall = \frac{TP}{(TP + FN)} \quad (6)$$

- 2) **Precision.** Precision assesses how many of the instances classified as positive are actually correct[16][21]. Accordingly, the formula for precision is defined as:

$$Precision = \frac{TP}{(TP + FP)} \quad (7)$$

- 3) **F1 Score.** The value is derived by taking the harmonic mean of both precision and recall and is used for binary, multiclass and multilabel classification problems[16][21]. It becomes 1 only when precision and recall are both 1. Hence, F1 score is calculated using the following formula:

$$F1 \text{ Score} = 2 * \frac{Precision * Recall}{Precision + Recall} \quad (8)$$

- 4) **Balanced Accuracy.** This metric is used for evaluating the classification models with imbalanced data. It is defined as the average of recall also known as macro-averaged recall[17][21]. For calculating balanced accuracy, the used formula is

$$TPR = \frac{TP}{(TP + FN)} \quad \text{and} \quad TNR = \frac{TN}{(TN + FP)} \quad (9)$$

$$Balanced \text{ Accuracy} = \frac{(TPR + TNR)}{2} \quad (10)$$

Chapter 4

Results and Discussions

4.1 Experimentation Results

4.1.1 Dropping the features

The original dataset contains 57 features out of which 29 features are eliminated as shown in the table 2 below. These features are dropped as they demonstrated either a high redundancy or lacked relevance with respect to the target variable. Furthermore, it was observed that 2 out of the 29 eliminated features contained approximately 80% of the missing values, thereby justifying their exclusion from the subsequent analysis. Figure 3 shows the correlation of numerical features.

Table 2: List of Features to be Dropped

Sr. No.	Feature Name	Sr. No.	Feature Name	Sr. No.	Feature Name
1	P_DETECTION	11	S_RA_STR	21	S_DEC_TXT
2	S_DISCOVERY_FACILITY	12	S_DEC_STR	22	S_LOG_LUM
3	P_YEAR	13	S_MAG	23	S_SNOW_LINE
4	P_UPDATE	14	S_DISTANCE	24	P_HABZONE_OPT
5	P_MASS_ORIGIN	15	S_LOG_G	25	P_HABZONE_CON
6	P_INCLINATION	16	P_POTENTIAL	26	P_TYPE_TEMP
7	P_OMEGA	17	P_DISTANCE	27	S_CONSTELLATION
8	S_NAME_HD*	18	P_TYPE	28	S_CONSTELLATION_ABR
9	S_NAME_HIP*	19	S_TYPE_TEMP	29	S_CONSTELLATION_ENG
10	S_TYPE	20	S_RA_TXT		

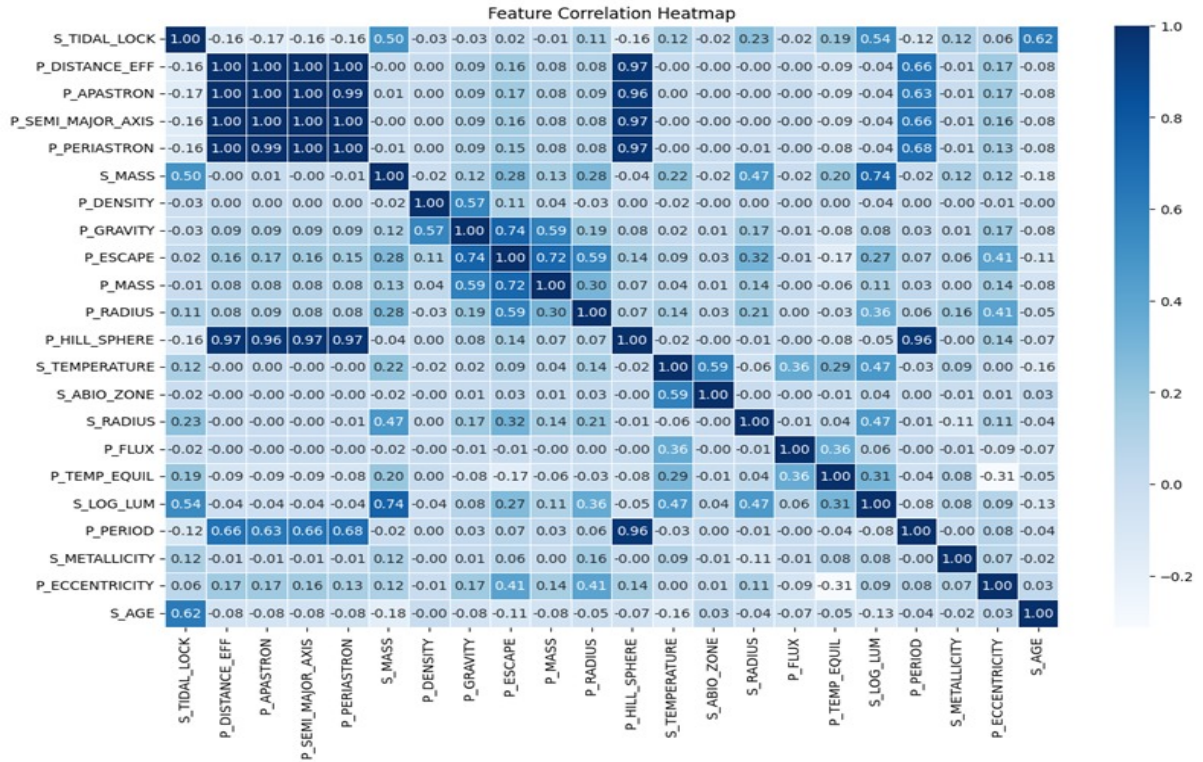


Figure 3: Feature correlation heatmap after dropping the redundant features

4.1.2 Datatype of features

Table 3 shows the value counts of the features before and after dropping the features.

Table 3: Datatype counts of features

Data Types	Before Dropping the Features	After Dropping the Features
float64	34	25
object	19	2
int64	4	1
Total	57	28

4.1.3 Feature Imputation

As there exists missing values in the dataset thus, different imputation methods are applied by considering the following aspects:

- 1) Outliers.
- 2) Distribution of the feature.
- 3) Variance Change.

Imputation methods namely, Median Imputation, KNN Imputation, Iterative Imputation, and XGBRegressor, are used to impute the features that contains missing values. Table 4 lists the features that are imputed using the median and Table 5 shows the features that are transformed before the imputation.

Table 4: Features Imputed Using Median Imputation

Sr. No.	Feature Name	Sr. No.	Feature Name
1	S_TIDAL_LOCK	7	P_DENSITY
2	P_DISTANCE_EFF	8	P_GRAVITY
3	P_APASTRON	9	P_ESCAPE
4	P_SEMI_MAJOR_AXIS	10	P_MASS
5	P_PERIASTRON	11	P_RADIUS
6	S_MASS	12	S_LUMINOSITY

Table 5: List of transformed features

Transformed features	
S_ABIO_ZONE	P_HILL_SPHERE
P_ECCENTRICITY	S_RADIUS
S_AGE	S_METALLICITY
S_TEMPERATURE	P_FLUX
P_TEMP_EQUIL	P_PERIOD

Table 6: Features Imputed Using Different Imputation Techniques

P_DISTANCE_EFF	S_TEMPERATURE	P_TEMP_SURF	S_AGE
P_APASTRON	S_METALLICITY	P_HILL_SPHERE	P_FLUX
P_SEMI_MAJOR_AXIS	S_ABIO_ZONE	S_RADIUS	P_PERIOD
P_TEMP_EQUIL	P_ECCENTRICITY	P_PERIASTRON	S_MASS

In table 5, the features are transformed as they are heavily skewed so direct distance-based imputation cannot be applied. Thus, it needs to be transformed first to make it normally distributed or the distribution must be close to normal distribution. After transformation, upon experimenting different imputation methods are applied based on the optimal values of the parameters as shown in Table 6.

4.2 Software and Hardware Requirements

Hardware:

- 8GB RAM
- 512GB ROM
- x64 Operating System / MacOS

Software:

- Python 3.12
- IDE : Google Colab
- Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, miceforest, xgboost, scipy, streamlit.

4.3 Results

4.3.1 Model performance with only three derived features

Table 7 describes the performance of Random Forest and XGBoost by using only three derived features namely Earth similarity index, atmospheric retention and long-term stability.

Table 7: Model's performance with derived features

Models	Precision	Recall	F1 Score	Balanced Accuracy
Random Forest	0.9845	0.9839	0.9839	0.6005
XGBoost	0.9804	0.9866	0.9835	0.4806

4.3.2 Model performance with all features (reduced and derived)

Table 8 describes the performance of Random Forest and XGBoost by using all features including reduced and derived ones.

Table 8: Model's performance with all features

Models	Precision	Recall	F1 Score	Balanced Accuracy
Random Forest	0.9954	0.9955	0.9955	0.8777
XGBoost	0.9982	0.9982	0.9981	0.9259

4.4 Discussions

In this study, the dataset is sourced from the NASA Exoplanet Archive, accessed on 12 March 2024, comprising 5599 entries with a total of 57 features. An initial exploratory data analysis is conducted to understand the structural characteristics and distributional properties of the dataset. The data types include 34 continuous (float64), 19 categorical (object), and 4 discrete (int64) features. Given the substantial presence of missing values, appropriate data preprocessing steps are essential prior to model development. 29 out of the 57 features are dropped since they are redundant or irrelevant with respect to the target variable. Thereafter, various imputation techniques—such as Median Imputation, K-Nearest Neighbors Imputation, Iterative Imputation, and XGBRegressor-based Imputation—are employed based on the nature and extent of missingness. For features exhibiting significant skewness, particularly those following log-normal or pareto distributions, transformation methods are applied to approximate normality before implementing the imputation strategies. Derived features such as the Earth Similarity Index, Atmospheric Retention, and Long-Term Stability are computed from existing attributes to encapsulate essential exoplanetary characteristics, thereby facilitating improved classification performance. Following the preprocessing phase, machine learning algorithms—namely, Random Forest and XGBoost—are implemented to classify the exoplanets. The models are initially trained using the complete set of features, and subsequently, using only the derived features, such as Earth Similarity Index, Atmospheric Retention, and Long-Term Stability. For the complete set of features, the Random Forest and XGBoost models achieved balanced accuracies of 87.77% and 92.59%, respectively. The performance evaluation metrics - Precision, Recall, and F1-Score - for Random Forest are 99.54%, 99.55%, and 99.55%, while XGBoost yielded 99.82%, 99.82%, and 99.81%, respectively. For the derived feature set comprising Earth Similarity Index (ESI), Atmospheric Retention (AR), and Long-Term Stability (LTS), Random Forest attained a balanced accuracy of 60.05%, and XGBoost achieved 48.06%. The corresponding Precision, Recall, and F1-Score for Random Forest were 98.45%, 98.39%, and 98.39%, while XGBoost recorded 98.04%, 98.66%, and 98.35%, respectively. Hence, it can be concluded that XGBoost demonstrates superior performance over the other implemented models when trained on the complete set of features, achieving a balanced accuracy of approximately 92.59%.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

As the exploration of potentially habitable exoplanets advances, creating robust predictive frameworks becomes increasingly essential. Initially, the dataset exhibited considerable missing values, which could compromise model accuracy and reliability. To address this, appropriate feature imputation techniques are applied to ensure data completeness. Following this, feature engineering led to the derivation of three scientifically meaningful features—Earth Similarity Index (ESI), Atmospheric Retention (AR), and Long Term Stability (LTS)—that align more closely with the real-world criteria used in habitability assessments and help reduce dimensionality. Subsequently, two tree-based classification models, Random Forest and XGBoost, are employed due to their robustness and independence from feature standardization. Among them, XGBoost demonstrated superior performance, achieving a balanced accuracy of 92.59% when all features (reduced and derived) are used, making it the most effective model for identifying potentially habitable exoplanets in this study. To make the research more useful and accessible, a web-based platform is proposed using Streamlit, a Python framework for building interactive web applications. This platform allows users to select data about an exoplanet to check how likely it is to support life. It also identifies the star the planet orbits and provides details about the star—such as its type, temperature, size, and age. The platform explains how these star features affect the planet’s chances of being habitable.

5.2 Future Work

Future work aims to adopt more empirical approaches to improve how accurately habitability is determined. One possible improvement is to study the absorption coefficient more closely, which could help correct its effect on the classification results. It is also important to explore how habitable zone planets are spread around different types of stars, to better understand how a star’s features affect the chances of a planet being habitable. Another important aspect is to look at the problems caused by current detection methods like the transit and radial velocity techniques. These methods can make it easier to find certain kinds of planets, which creates an unfair view of what kinds of planets are out there. This bias affects how habitability is understood. Reducing these detection issues will lead to a more accurate and balanced view of exoplanet habitability.

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